

CURRICULUM VITAE

OLEG GENDELMAN

January, 2026

Date and place of birth: 18.07.1969, Kharkov, Ukraine

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ACADEMIC DEGREES

2000 – **Doctor of Sciences**, Physics and Mathematics, from the Higher Commission of Attestation, Russian Ministry of Education, thesis name: “Investigation of Structural Defects and Localized Excitations in Polymer Crystals and Glasses by Methods of Nonlinear Dynamics”

1995 - **Ph.D.**, Physics and Mathematics, from N.N.Semenov Institute of Chemical Physics RAS

1992 – **M Sc**, Applied Physics and Mathematics, Moscow Institute of Physics and Technology, Department of Molecular and Chemical Physics

ACADEMIC APPOINTMENTS

2015 - now : Samuel and Anne Tolkowsky Professor, Faculty of Mechanical Engineering, Technion – Israel Institute of Technology, Haifa, Israel

2014 – 2015: Professor with tenure, Faculty of Mechanical Engineering, Technion – Israel Institute of Technology, Haifa, Israel

2007 - 2013: Associate Professor with tenure, Faculty of Mechanical Engineering, Technion – Israel Institute of Technology, Haifa, Israel

2003 - 2007: Senior Lecturer, Faculty of Mechanical Engineering, Technion – Israel Institute of Technology, Haifa, Israel

2002 - 2003: Leading Research Fellow, N.N.Semenov Institute of Chemical Physics RAS, Department of Polymer and Composite Materials

2000 - 2002: Senior Research Fellow, N.N.Semenov Institute of Chemical Physics RAS, Department of Polymer and Composite Materials

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1995 – 2000: Research Fellow, N.N.Semenov Institute of Chemical Physics RAS,
Department of Polymer and Composite Materials

1992 – 1995: Graduate student, Moscow Institute of Physics and Technology

RESEARCH INTERESTS

Applied Mathematics, Nonlinear dynamics, nonlinear oscillations in discrete and continuous systems, nonlinear normal modes, energy transfer, vibration protection and mitigation, dynamics and transport phenomena in low-dimensional systems, and applications to polymer systems and granular materials

TEACHING EXPERIENCE

Linear Systems – undergraduate;

Dynamics – undergraduate

Analytic Methods in Mechanical Engineering 1 –undergraduate & graduate

Analytic Dynamics – undergraduate & graduate

DEVELOPMENT OF COURSES AND ACADEMIC PROGRAMS

2021 – Development of new modular undergraduate program for the Faculty (approved and works since 2021).

2005 – course 038801 - Models of Nonlinear Dynamics - graduate

TECHNION ACTIVITIES

Since 2016 – Member of Technion Senate

2013 –2016 Technion Academic Development Committee

Since 2004 - Technion Committee on Evaluation of Candidates from the Former Soviet Union

Since 2007 - Technion Interdisciplinary Committee on Polymer Engineering

DEPARTMENTAL ACTIVITIES

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2025 – Vice Dean on Academic Development, Faculty of Mechanical Engineering

2019 – 2022 Dean of Faculty of Mechanical Engineering

2017 – 2018 Vice – Dean on Graduate Studies

2010 – 2012 Undergraduate Studies Coordinator

2008 - 2009 Coordinator of Excellence Programs ("Brakim" and "Reamim")

2004 – 2007 Faculty Seminar Coordinator

PUBLIC PROFESSIONAL ACTIVITIES

Member of Editorial Board

Scientific Reports – since 2016

Acta Mechanica – since 2025

HONORS

2018 – Cooper Award for Excellence in Research, Technion

2016 – “Person of the year” award in the field of science, 9th Channel of Israel TV.

2016 – Outstanding Referee award, American Physical Society

2015 - Samuel and Anne Tolkowsky Chair in Mechanical Engineering

2012 – Water Arbitration Prize, Institution of Mechanical Engineers, London, UK , for the best paper published in 2011.

2011 – Award for Outstanding Contribution, Brakim program, IDF

2010 – Included on list "Outstanding Immigrant Scientists" by Ministry of Absorption.

2006 - Evelyn and Salman Grand Academic Lectureship, Technion, Israel

2003 – Horev Fellow, Shalom and Taub foundations, Program “Leaders in Science and Technology”, Technion, Israel

2001 - Acting Member, Russian Academy of Natural Sciences

2001 – Outstanding Researcher Stipend, Russia Academy of Sciences

2000 – Young Scientist Fellowship, Russian Academy of Sciences

1998 - State Fellowship for Outstanding Researchers

1994 - Soros Graduate Student Award

MEMBERSHIP IN PROFESSIONAL SOCIETIES

SIAM, ASME, EUROMECH

POSTDOCTORAL ASSOCIATES

1. Dr. Avramov Konstantin, 2006
2. Dr. Serov Alexander, 2007
3. Dr. Strozzi Matteo, 2018
4. Dr. Kravets Pavel, 2019
5. Dr. Farid Maor, 2020
6. Dr. Majdi Gzal, 2022

SUPERVISION OF GRADUATE STUDENTS

Theses Completed – PhD (9)

1. Starosvetsky Yuli – primary supervisor, 2009, “Strongly Nonlinear Vibration Absorber” (Assistant Professor, Faculty of Mechanical Engineering, Technion). Pnueli Prize, 2009
2. Zolotarevskiy Vadim – primary supervisor, 2016, “Heat Transport in Low-dimensional Models: Effects of Disorder and Dimensionality” (co-supervised with Asst. Prof. Y. Starosvetsky).
3. Farid Maor – primary supervisor, 2017, “Nonlinear Liquid Sloshing in Partially-Filled Tanks: Modelling, Exploration and Mitigation”
4. Grinberg Itay – primary supervisor, 2017, “Localization and Energy Transport in Vibro-Impact Systems”.
5. Perchikov Nathan – primary supervisor, 2019, “Nonlinear Dynamics of Discrete Mechanical Systems with Flat Dispersion Bands.
6. Shiroky Itzhak – primary supervisor, 2020, “Front propagation in bi-stable non-degenerate systems: model dependence and universality”.

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7. Paul Jithu – primary supervisor, 2021, “Kapitza resistance in linear and nonlinear chain models”
8. Gazal Majdi – primary supervisor, 2022, “Stationary and Transient Nonlinear Processes in Engineering Systems”
9. Karmi Gleb – primary supervisor, 2022, “Analytic Exploration of Safe Basins in a Benchmark Problem of Forced Escape”

Theses Completed – MSc (35)

1. Starosvetsky Yuli – primary supervisor, 2006, “Optimization of Strongly Nonlinear Vibration Absorber” (continued to PhD studies). Barazani Prize, 2006
2. Shiroky Itzhak - primary supervisor, 2008, "Parametrically Excited Oscillator with Nonlinear Energy Sink" (“Brakim” student program¹, continued to PhD studies]
3. Meimukhin Danila – primary supervisor, 2008, “Systems with Strongly Non Linear Attachment Under Periodic Impact Loading” (PhD student at the Faculty).
4. Shvartsman Rina – primary supervisor, MSc, 2010, "Non-Fourier Heat Conduction in Microscopic Models of Dielectrics" (Continues PhD studies in Australia).
5. Bar Tamir – primary supervisor, 2010, “Suppression of Limit Cycle Oscillations with the help of Strongly Nonlinear Attachment”. (“Brakim” student program, currently: officer, IDF)
6. Dubrovsky Alexander – co – supervisor (primary - Dr. M.Shusser), "Mass Transfer Controlled Bubble Growth in Concentrated Polymer Solutions", 2010, (PhD student at the Faculty).
7. Madar Baruh – primary supervisor, 2010 "Suitability of continuous Equations of Heat Conduction in Microscopic Models of Dielectrics", (external, Iscar).
8. Shalev Ariel - primary supervisor, 2010 " Recoiling of weapon in moving vehicles", (“Brakim” student program, currently: officer, IDF).

¹“Brakim” program is a special BSc and MSc program for outstanding students, trained for leading RnD positions in Israel Defense Force.

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9. Shuster Guy Sergei –primary supervisor (together with Prof. Y. Ben-Haim), 2011, “Design of suspension system for an off-road vehicle under uncertainty of the terrain” (“Brakim” student program, currently: officer, IDF).
10. Kachman Tal – primary supervisor, 2011 , “ Nonstationary heat conduction in disordered lattices”
11. Kedem Shelli - primary supervisor, 2011 topic: "Dynamics of Localization in Systems with Neutral Delay Coupling" (“Brakim” student program, currently: officer, IDF).
12. Edelman Kobi - primary supervisor, 2012, topic: "Dynamics of Self - Excited oscillators with Neutral Delay Coupling" (Rafael)
13. Avraham Ofir - primary supervisor, 2012, topic: "Nonlinear phenomena in Hydraulic Vibration Absorbers" (“Brakim” student program, currently: officer, IDF)
14. Ramus Artyom – co-supervisor (together with Dr. M.Shusser), 2013, topic: “Laminar Flow in a Curved Pipe with a Sudden Expansion”
15. Domany Elad - primary supervisor, 2013, topic: “Mitigation of undesired limit cycle oscillations in self – excited nonlinear systems” (“Brakim” student program, summa cum laude, Barazani prize).
16. Benarous Nir – primary supervisor, 2014, topic: “Nonlinear Energy Sink with Combined Nonlinearities” (“Brakim” student program).
17. Veremkroit Michael - primary supervisor, 2014, topic: “Analytic Exploration of Discrete Breathers in a Forced-Damped Klein-Gordon Type Chain” (“Brakim” student program)
18. Halioua Guy – primary supervisor, 2014, topic: “Waves of Collapse in Nondegenerate Chain Arrays” (“Brakim” student program)
19. Uzan Nissan - primary supervisor, 2015, topic “Quenching and synchronization in systems of phase-only oscillators with time delay” (“Brakim” student program).
20. Erez Ariel - primary supervisor, 2015, topic: “Dynamics of Van Der Pol Oscillator with Rotational Energy Sink” (external)

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21. Karmi Gleb – primary supervisor, 2015, topic: “Bouncing Ball Inside the Vibrating Circular Fence Chaotic System Analysis” (external)
22. Farid Maor – primary supervisor, 2015, topic: “Tuned Pendulum as Nonlinear Energy Sink for Broad Energy Range” (“Brakim” student program)
23. Alloni Aviv – primary supervisor, 2015, topic: “Dynamics of Forced System with Vibro Impact Energy Sink” (“Brakim” student program)
24. Slavkin Ilya – primary supervisor, 2015, topic: “High Frequency Nonlinear Micro Mass Sensor of Enhanced Performance and Sensitivity” (“Brakim” student program)
25. Degtyar Andrey – primary supervisor, 2018, topic "Flexible and Adaptable Ankle Foot Orthosis for Walking Style Correction of Post Stroke Patients".
26. Naiger Dan – primary supervisor, 2019, "Escape of a Forced-Damped Particle from a Potential Well: Transient Response".
27. Ezra Tal - primary supervisor, 2019, "Escape of excited two DOF system from potential well".
28. Shaban Noa – primary supervisor, 2019, "Non-reciprocity in Non-Dispersive Waveguides with Strongly Nonlinear Coupling"
29. Tzemah Naor – primary supervisor, 2020, "Accelerating oscillatory fronts in a chain with nonlocal interactions: effect of linear spectrum"
30. Zussman Dar – primary supervisor, 2020, "Effect of finite vessel stiffness on transition from two-dimensional liquid sloshing to swirling: reduced-order modeling".
31. Moskovitch Omer – primary supervisor, 2021, “Resonance and Energy Transfer in Forced Vibro-Impact Systems with Compliance”
32. Bader Alla – primary supervisor, 2021, “Supratransport in a Vibro-Impact Oscillatory Chain”
33. Hoffman Yuval – primary supervisor, 2022, “Transition to 3D Motion in an Equivalent Mechanical Model of Nonlinear Liquid Sloshing”
34. Engel Amit – primary supervisor, 2023, “Escape of Two DOF Dynamical System from the Potential Well”

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35. Veltman Yuval – primary supervisor, 2024, “Intermodal targeted energy transfer in two-dimensional setting”.

Theses in progress – PhD

1. Varshavchik Evgeniy – started in 2023
2. Chuanqi Wei – started in 2024, GTIIT

Theses in Progress – MSc

1. Kanciper Yuval – started in 2023
2. Wolfovich Yuval – started in 2023
3. Avitan Ori – started in 2023
4. Sokolsky Daniel – started in 2024
5. Aksenton Yam – started in 2024
6. Kotser Sarit – Started in 2024

RESEARCH GRANTS

2025 – 2030 – **Israel Science Foundation**, grant 3085/25, “Escape dynamics: inside and beyond the resonance”, PI, NIS 220,000 for 2025-2026.

2023-2025 – **Deutsche Forschungsgemeinschaft (DFG)**, grant 508244284, “Escape dynamics in Engineering Systems”, PIs: Prof. A. Fidlin (Karlsruhe Institute of Technology), Prof. O.V.Gendelman, Euro 116,250 (for all period).

2021-2025– **Israel Science Foundation**, grant 2598/21, “Intermodal Targeted Energy Transfer.”, PI, NIS 230,000 for 2021-2022.

2020 – 2021, **MISTI Foundation, MIT-Israel Zuckerman STEM Fund**, Transient strongly nonlinear dynamics: opportunities and challenges, PIs: Prof. Themistoklis Sapsis, Prof. O.V.Gendelman, \$25,000.

2017-2021, **PAZY Foundation**, grant 298/18, "Liquid Sloshing in Seismically Excited Tanks: Prediction and Mitigation of Structural Damage", NIS 209,000 for 2017-2018, co-PI: Dr. Eilon Shimshi, CI: Prof. Pinhas Bar-Yoseph.

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2017-2021– **Israel Science Foundation**, grant 1696/17, “Energy exchanges, resonances and waves in essentially nonlinear systems.”, PI, NIS 230,000 for 2017-2018.

2013-2017 – **Israel Science Foundation**, grant 838/13, “Localization in Forced/Damped Strongly Nonlinear Systems”, PI, NIS 170,000 for 2013.

2009–2013 – **Binational Science Foundation (BSF)**, grant 2008055 – Nonlinear Dynamics of Oscillators Exhibiting Targeted Energy Transfer, \$33,000 for 2009, other PIs: Professor Alexander Vakakis, Professor Lawrence Bergman, University of Illinois, Urbana – Champaign

2008 – **Royal Society of Edinburgh**, joint research with Prof. M. Wiercigroch, University of Aberdeen

2006-2009 – **Israel Science Foundation**, grant 486/05 – “Strongly Nonlinear Vibration Absorber”, PI, NIS180.000 for 2006.

1997-2002 – Russian Foundation of Basic Research (participant of 11 research grants with Profs. L. Manevitch, A.A. Berlin, A.V.Savin, V.G. Oshmyan and L.A.Novokshonova as PIs, 6 travel grants for participation in international conferences).

2001 – 2003 Air Force Office of Scientific Research. Contract 00-AF-B/V-0813 (Dr. Dean Mook is the Grant Monitor, PIs – Prof. Alexander Vakakis and Prof. Larry Bergman, University of Illinois in Urbana - Champaign), topic : “Concept of energy sink and strongly nonlinear vibration absorbers”, personal funding - \$25.000.

2001 – Young Researcher Grant (6th Competition, grant No. 123), Russia Academy of Sciences, Principal Investigator (group of 7 participants), topic: “Dynamical description of transfer processes in polymer crystals”

PATENTS

A.Degtyar and **O.V.Gendelman**, FOOT ORTHOSIS, US12,239,561 B2, granted March 4, 2025

PUBLICATIONS

Theses

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1. “Investigation of Structural Defects and Localized Excitations in Polymer Crystals and Glasses by Methods of Nonlinear Dynamics”, DSc thesis, Institute of Chemical Physics RAS, 1999
2. Structural Defects and Plasticity in Low-molecular and Polymer Glasses”, PhD thesis, Institute of Chemical Physics RAS, 1995, supervisor – Prof. L.I. Manevitch

Refereed papers in professional journals

Published papers

Single – authored papers are printed boldface. In the papers published with supervised students and postdocs, their names are underlined.

1. **O.V. Gendelman** and L.I. Manevitch, Nonlinear dynamics of a diatomic Toda lattice and heat-conduction problem in quasi-one-dimensional crystals, *Sov. Phys. JETP*, v.102(2), 511-521, 1992
2. **O.V. Gendelman** and L.I. Manevitch, New model of plastic deformation of disordered systems. *Journal of Physics: Condensed Matter*, v.5, 1633-1642, 1993
3. **O.V. Gendelman** and L.I. Manevitch, Linear and Nonlinear Excitations in Polyethylene Crystal, *Zhurnal. Fizicheskoi Khimii*, v.69, 57-61, 1995
4. N.K. Balabaev, **O.V. Gendelman**, M.A. Mazo and L.I. Manevitch, Molecular Dynamics Modeling of Essentially Nonlinear Phenomena in Polyethylene Crystals *Zhurnal Fizicheskoi Khimii*, v.69, 24-27, 1995
5. **O.V. Gendelman** and L.I. Manevitch, A model of plastic deformation and localized vibration modes in 3D glass, *Journal of Physics: Condensed Matter*, v.7, .6993-7004, 1995
6. **O.V. Gendelman** and L.I. Manevitch, The description of polyethylene crystal as a Continuum with internal degrees of freedom. *International Journal of Solids and Structures*. v.33, 1781-1798, 1996
7. N.K. Balabaev, **O.V. Gendelman** and L.I. Manevitch, Self-assembly of domain wall of molecular twist defects in polyethylene crystal, *Macromolecular Symposia*, v.106, 31-39, 1996.

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8. **O.V. Gendelman** and L.I. Manevitch, Structural Defects and Low-Frequent Localized Modes in Disordered Systems, *Zhurnal Experimentalnoi i Teoreticheskoi Fiziki*, v. 110(1), 287-297, 1996
9. N.K.Balabaev, **O.V. Gendelman**, M.A.Mazo and L.I. Manevitch, Modeling Twist Domain Walls in Polyethylene Crystals, *Vysokomolekylarnye Soedineniya, ser. A*, v.38, 676-681, 1996.
10. **O.V. Gendelman** and L.I. Manevitch, Exact Soliton-Like Solutions in Generalized Dynamical Models of a Quasi-One-Dimensional Crystal, *Zhurnal Experimentalnoi i Teoreticheskoi Fiziki*, 85(4), 824-826, 1997
11. **O.V. Gendelman** and L.I. Manevitch, Linear and nonlinear excitations in a polyethylene crystal. 1. Vibration modes and linear equations *Macromolecular Theory and Simulations*, v.7, 579-589, 1998
12. **O.V. Gendelman** and L.I. Manevitch, Linear and nonlinear excitations in a polyethylene crystal. 2 Nonhomogeneous states and nonlinear excitations, *Macromolecular Theory and Simulations* v.7, 591-598, 1998
13. **O.V. Gendelman** and L.I. Manevitch, Asymptotic Study of Damped 1D Oscillator with Close to Impact Potential, in: *Dynamics of Vibro-Impact Systems*, Springer, ed. V.I. Babitsky, 159-166, 1998
14. A.A. Berlin, **O.V. Gendelman**, N.N. Sinelnikov, M.A. Mazo and L.I. Manevitch, The Analysis of Mechanical and Thermodynamical Properties of Binary System of Disks Depending on their Ordering, *Doklady Akademii Nauk*, v.361, 779-783, 1998
15. A.A. Berlin, **O.V. Gendelman**, N.N. Sinelnikov, M.A. Mazo and L.I. Manevitch, On Solid-Liquid Transition in plane disk systems, *Journal of Physics: Condensed Matter*, v.11, 4583-4596, 1999
16. A.A. Berlin, N.N. Sinelnikov, **O.V. Gendelman**, M.A. Mazo and L.I. Manevitch, On the Solid-Liquid transition in the system of disks on a plane, *Biofizika*, v.44, 953-955, 1999
17. G. Salenger, A.F. Vakakis, **O.V. Gendelman**, I.V. Amdrianov and L.I. Manevitch, Transitions from strongly- to weakly-nonlinear motions of damped nonlinear oscillators, *Nonlinear Dynamics*, v.20, 99-114, 1999

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18. A.V.Savin and **O.V. Gendelman**, Torsion Solitons in Linear Macromolecules, *Vysokomolekylarnye Soedineniya, Ser.A*, v.41, .263-270, 1999
19. **O.V. Gendelman** and A.F. Vakakis, Transition from Localization to Nonlocalization in Strongly Nonlinear Damped Oscillators, *Chaos, Solitons and Fractals*, v. 11, 1535-1542, 2000
20. **O.V. Gendelman** and L.I. Manevitch, Reflection of Short Rectangular Pulses in the ideal string attached to strongly nonlinear oscillator, *Chaos, Solitons and Fractals*, v. 11, .2473-2477, 2000
21. A.A. Berlin, N.N. Sinelnikov, **O.V. Gendelman**, M.A.Mazo and L.I. Manevitch, An analysis of the structure and thermodynamic properties of bicomponent systems of disks and spheres, *Russian Journal of Physical Chemistry*, v.74, S46-S51, 2000
22. **O.V. Gendelman** and A.V.Savin, Normal heat conductivity of the one-dimensional lattice with periodic potential of nearest-neighbor interaction, *Phys. Rev. Letters*, v. 84, 2381-2384, 2000
23. **O.V. Gendelman**, K.E.Kuporosoov and L.I. Manevitch, The Formation of Soliton-type Nonlinear Excitations During Heat Transfer in a Crystalline Poly(ethylene) Chain, *Polymer Science, ser. A*, v.42, 1337-1345, 2000
24. A.V.Savin and **O.V. Gendelman**, On the finite thermal conductivity of a one – dimensional rotator lattice, *Physics of the Solid State*, v.43, 355-364, 2001
25. **O.V. Gendelman** **Transition of Energy to a Nonlinear Localized Mode in a Highly Asymmetric System of Two Oscillators**, *Nonlinear Dynamics*, v.25, 237-253, 2001
26. **O.V. Gendelman**, A.F. Vakakis, L.I. Manevitch and R. McCloskey, Energy Pumping in Nonlinear Mechanical Oscillators I: Dynamics of the Underlying Hamiltonian System, *Journal of Applied Mechanics – Transactions of the ASME*, v.68, 34-41, 2001
27. A.F. Vakakis and **O.V. Gendelman**, Energy Pumping in Nonlinear Mechanical Oscillators II: Resonance Capture, *Journal of Applied Mechanics – transactions of ASME*, v.68, 42-48, 2001

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28. V.V. Ginzburg, **O.V. Gendelman** and L.I. Manevitch, Simple “Kink” Model of Melt Intercalation in Polymer-Clay Nanocomposites, *Physical Review Letters*, v.86, 5073-5075, 2001
29. N.K. Balabaev, **O.V. Gendelman** and L.I. Manevitch, Supersonic Motion of Vacancies in a Polyethylene Crystal, *Physical Review E*, v.64, 036702 (1-8), 2001
30. A.A. Berlin, N.N. Sinelnikov, **O.V. Gendelman**, M.A.Mazo and L.I. Manevitch, Melting of Crystals Composed of Elastic and Lennard – Jones Particles, *Doklady Physical Chemistry*, v.382, 66-69, 2002
31. A.F. Vakakis, L.I. Manevitch, **O. Gendelman** and L. Bergman, Dynamics of Linear Discrete Systems Connected to Local Essentially Nonlinear Attachments. *Journal of Sound and Vibration*, v.264, 559-577, 2003.
32. L.I. Manevitch, **O. Gendelman**, A. I. Musienko, A. F. Vakakis and L.Bergman, Dynamic Interaction of a Semi-infinite Linear Chain of Coupled Oscillators with a Strongly Nonlinear End Attachment, *Physica D*, v.178, 1-18, 2003
33. A. V. Savin and **O. V. Gendelman**, Heat conduction in one-dimensional lattices with on-site potential, *Physical Review E*, v.67, 041205, 2003
34. **O.V. Gendelman**, L.I. Manevitch and O.L. Manevitch, Solitonic Mechanism of Structural Transition in Polymer-Clay Nanocomposites, *Journal of Chemical Physics*, v.119, 1066-1069, 2003
35. **O. Gendelman**, L. I. Manevitch, A. F. Vakakis and L. Bergman, A Degenerate Bifurcation Structure in the Dynamics of Coupled Oscillators with Essential Stiffness Nonlinearities, *Nonlinear Dynamics*, v.33, 1-10, 2003
36. A F. Vakakis, D. M McFarland, L. Bergman, L I. Manevitch and **O. Gendelman**, Isolated Resonance Captures and Resonance Capture Cascades Leading to Single- or Multi-Mode Passive Energy Pumping in Damped Coupled Oscillators, *Journal of Vibration and Acoustics – Transactions of the ASME*, v. 126, 235-244, 2004
37. **O.V. Gendelman** and A.V.Savin, Heat Conduction in a One-Dimensional chain of Hard Discs with Substrate Potential, *Physical Review Letters*, v.92, 074301, 2004

38. **O.V. Gendelman**, **Bifurcations of Nonlinear Normal Modes of Linear Oscillator with Strongly Nonlinear Damped Attachment**, *Nonlinear Dynamics*, v.37, 115-128, 2004.
39. A.A.Berlin., **O.V. Gendelman**, M.A. Mazo, L.I. Manevitch, Thermal Expansion Coefficient in Simple Models of Condensed Media, *Doklady: Physical Chemistry*, v.397, 187-190, 2004
40. **O.V. Gendelman** and C.-H. Lamarque, Dynamics of Linear Oscillator Coupled to Strongly Nonlinear Attachment with Multiple States of Equilibrium, *Chaos, Solitons and Fractals*, v.24, 501-509, 2005
41. E. Bormashenko, R. Pogreb, O. Stanevsky, Y. Bormashenko, T. Stein, V.-Z. Gaisin, R. Cohen and **O.V. Gendelman**, Mesoscopic Patterning in Thin Polymer Films Formed under the Fast Dip-Coating Process, *Macromolecular Materials and Engineering*, v.290, 114-121, 2005
42. E. Bormashenko, R. Pogreb, O. Stanevsky, Y. Biton, Y. Bormashenko, Y. Socol and **O. Gendelman**, Self-Assembled Honeycomb Polycarbonate Films Deposited on Polymer Piezoelectric Substrates and their Applications, *Polymers for Advanced Technologies*, v.16, 209-304, 2005
43. **O.V. Gendelman**, D.V. Gorlov, L.I. Manevitch and A.I. Musienko, Dynamics of coupled linear and essentially nonlinear oscillator with substantially different masses, *Journal of Sound and Vibration*, v.286, 1-19, 2005
44. **O.V. Gendelman** and A.V. Savin, Reply to Comment on "Normal heat conductivity of the one-dimensional lattice with periodic potential of nearest-neighbor interaction", by L.Yang, B.B.Hu, *Physical Review Letters*, v.94, 219405 Jun 3, 2005
45. E. Bormashenko, R. Pogreb, O. Stanevsky, Y. Bormashenko, S. Tamir, R. Cohen, M. Nunberg, V.-Z. Gaisin, M. Gorelik and **O.V. Gendelman**, Mesoscopic and submicroscopic patterning in thin polymer films: impact of the solvent, *Materials Letters*, v. 59, 2461-2464, 2005

46. E. Bormashenko, R. Pogreb, O. Stanevsky, Y. Bormashenko and **O. Gendelman**, Formation of honeycomb patterns in evaporated polymer solutions: Influence of the molecular weight, *Materials Letters*, v. 59, 3553 – 3557, 2005
47. E. Bormashenko, R. Pogreb, O. Stanevsky, Y. Bormashenko, T. Stein and **O.V. Gendelman**, Mesoscopic patterning in evaporated polymer solutions: new experimental data and physical mechanisms, *Langmuir*, v.21, 9604-9609, 2005
48. **O.V. Gendelman**, Modeling of Inelastic Impacts with the Help of Smooth Functions, *Chaos, Solitons and Fractals*, v. 28, 522–526, 2006
49. **O.V. Gendelman**, Degenerate Bifurcation Scenarios in the Dynamics of Coupled Oscillators with Symmetric Nonlinearities, *International Journal of Bifurcations and Chaos*, v. 16, 169-178, 2006
50. E. Bormashenko, R. Pogreb, O. Stanevsky, Y. Bormashenko, T. Stein, R. Cohen, Sh. Reis and **O.V. Gendelman**, Mechanisms of mesoscopic patterning in evaporated polymer films deposited on tilted and vertical substrates, *Journal of Materials Science*, v.41, 455-461, 2006
51. E. Bormashenko, R. Pogreb, A. Musin, O. Stanevsky, Y. Bormashenko, G. Whyman, **O. Gendelman** and Z. Barkay, Self-assembly in Evaporated Polymer Solutions: Influence of the Solution Concentration, *Journal of Colloid and Interface Science*, v.297, 534-540, 2006
52. M. Shusser and **O.V. Gendelman**, Stability of an Evaporating Thin Polymer Film, *International Communications in Heat and Mass Transfer*, v.33, 564-570, 2006
53. **O.V. Gendelman**, E. Gourdon and C.-H. Lamarque, Quasiperiodic Energy Pumping in Coupled Oscillators under Periodic Forcing, *Journal of Sound and Vibrations*, v.294, 651-662, 2006
54. **O.V. Gendelman**, M. Shapiro, Y. Estrin, R.J. Hellmig and S. Lekhtmakher, Grain size distribution and heat conductivity of copper processed by equal channel angular pressing, *Materials Science and Engineering A*, v.434, 88-94, 2006
55. D. Meimukhin and **O.V. Gendelman**, Response Regimes of Integrable Damped Strongly Nonlinear Oscillator under Impact Periodic Forcing, *Chaos, Solitons and Fractals*, v.32, 405-414, 2007

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56. A. Arinstein, M. Burman, **O. Gendelman**, and E. Zussman, Effect of supramolecular structure on polymer nanofibre elasticity, *Nature Nanotechnology*, v.2, 59-62, 2007
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1. O.V. Gendelman, L.I. Manevitch Normal Heat Conductivity of the Diatomic Toda Lattice, XVIII ICTAM, Haifa, Israel, 1992, Book of Abstracts.
2. O.V. Gendelman, L.I. Manevitch Localized Vibration Modes in Polymer Glasses, 10th European Polymer Spectroscopy Conference, St. Petersburg, October, 1993
3. O.V. Gendelman, L.I. Manevitch Asymptotic Methods in the Theory of Polymer Crystals, Int. Conference: Asymptotics in Mechanics, St. Petersburg, August, 1994, Book of Abstracts
4. O.V. Gendelman, L.I. Manevitch, Microscopic Mechanism of Plastic Deformation in 3D Disordered Systems, 2nd European Solid Mechanics Conference, Genoa, September 9-12, 1994

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6. N.K.Balabaev, O.V. Gendelman, M.A. Mazo, and L.I. Manevitch Essentially nonlinear effects in statics and dynamics of polyethylene crystal: analytical study and numerical simulation Intern. Conf. Nano-structures and self-assemblies in polymer systems, St.-Petersburg - Moscow, May 1995.
7. N.K. Balabaev, O.V. Gendelman, M.A. Mazo, L.I. Manevitch Self-assembly of domain wall of molecular twist defects in polyethylene crystal Intern. Conf. Nano-structures and self-assemblies in polymer systems, St.-Petersburg - Moscow, May 1995, p. OL65
8. O.V. Gendelman, L.I. Manevitch On the mechanism of plastic Deformation of simple and polymeric glasses, Intern. Conf. Nano-structures and self-assemblies in polymer systems, St.-Petersburg - Moscow, May 1995, p. OL63
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16. O.V. Gendelman, Approximate Solution of Generalized Nonlinear Evolution Equations by Means of Pade Approximations, International Congress on Mathematical Physics, London, 2000
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 25. O.V. Gendelman, Degenerate Bifurcation Scenarios in the Dynamics of Coupled Oscillators with Symmetric Stiffness Nonlinearities, *The 30th Israeli Conference on Mechanical Engineering*, Tel Aviv, 2005
 26. O.V. Gendelman, Modeling of Inelastic Impacts with the Help of Smooth Functions, *The 30th Israeli Conference on Mechanical Engineering*, Tel Aviv, 2005
 27. O.V. Gendelman, Yu. Starosvetsky, Quasiperiodic Response Regimes of Linear Oscillator Coupled to Nonlinear Energy Sink Under Periodic Forcing, *The 30th Israeli Conference on Mechanical Engineering*, Tel Aviv, 2005
 28. D. Quinn, G. Kerschen, A. Vakakis, O. Gendelman, L. Bergman, and T. Sapsis, Energy Transition from a Linear Oscillator to an Attached Mass through an Essential Nonlinearity, Eleventh Conference on Nonlinear Vibrations, Stability, and Dynamics of Structures, Blacksburg, VA USA, August 13–17, 2006
 29. O.V. Gendelman, Y. Starosvetsky, Attractors of Harmonically Forced Linear Oscillator with Attached Nonlinear Energy Sink, Eleventh Conference on Nonlinear Vibrations, Stability, and Dynamics of Structures, Blacksburg, VA USA, August 13–17, 2006
 30. O.V. Gendelman, Targeted Energy transfers in a system with soft nonlinearity, EUROMECH – 483, Porto, Portugal, July 9-11, 2007
 31. Y. Starosvetsky and O.V. Gendelman, Response Regimes of Strongly Nonlinear Vibration Absorber, EUROMECH – 483, Porto, Portugal, July 9-11, 2007
 32. O.V. Gendelman and Tamir Bar, Targeted Energy Transfer and Quasiperiodic Response Regimes in Van – Der – Pol Oscillator with Attached Nonlinear Energy Sink, 7th EUROMECH Solid Mechanics Conference, Lisbon, September 7-11, 2009

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33. O.V. Gendelman, Nonlinear Normal Modes in Systems with Time Delay, 3rd International Conference on Nonlinear Normal Modes and Localization in Vibrating Systems, September 26 - 30, 2009, Roma, Italy
34. O.V. Gendelman, Nonstationary heat conduction in one-dimensional models with substrate potential, International Congress of Mathematical Physics, August 5-11, 2012, Aalborg, Denmark.

Seminar Talks presented at

Institute of Chemical Physics RAS, Moscow, Russia – 2004, 2998
Technion – Israel Institute of Technology, Israel – 2002, 2005, 2007
University of Illinois in Urbana – Champaign, USA – 1998, 1999, 2003, 2009, 2010
University of Beer – Sheva, Israel - 2003
University of Tel – Aviv, Israel - 2006
Illinois Institute of Technology, Chicago, USA - 2003
National Technical University of Athens, Greece – 2004, 2006
Kharkov Polytechnic University, Ukraine – 2003, 2009
University of Aberdeen, Scotland - 2008
University Center of Judea and Samaria – 2005, 2008, 2010
Cornell University, 2011
Michigan State University, East Lansing, 2012
University of Michigan, Ann Arbor, 2012
Future University, Hakodate, Japan, 2012
Hebrew University of Jerusalem, Israel, 2012
South Illinois University, Edwardsville, 2013
Massachusetts Institute of Technology, 2014
Center For Theoretical Physics of Complex Systems, Daejeon, South Korea, 2015
Griffith University, Brisbane, 2019

SPECIAL PROFESSIONAL ACTIVITIES

Member of Organizing and Scientific Committees:

- 2nd International Conference on Nonlinear Normal Modes and Localization in Vibrating Systems, June 19-23, 2006, Samos, Greece
- NPPS-2008, International Conference on Nonlinear Phenomena in Polymer Solids and Low – Dimensional Systems, July 7 – 10, 2008, Moscow, Russia
- 3rd International Conference on Nonlinear Normal Modes and Localization in Vibrating Systems, September 26 - 30, 2009, Roma, Italy
- 3rd International Conference "Nonlinear Dynamics – 2010", September 21-24, Kharkov, Ukraine
- Permanent Committee, International Conference on Vibration Problems, since 2011
- International Conference on Vibration Problems, Prague, September 4-9, 2011
- 2nd International Symposium "Rare Attractors and Rare Phenomena in Nonlinear Dynamics RA'11", May 16 - 20, 2011, Riga - Jurmala, Latvia
- **Chairman of Organizing and Scientific Committees, 4th International Conference on Nonlinear Normal Modes and Localization in Vibrating Systems, July 1-5, 2012, Faculty of Mechanical Engineering, Technion, Israel (40 participants from 11 countries)**
- The 4th International Conference “Nonlinear Dynamics – 2013”, June 19-22, 2013, Sevastopol, Ukraine
- Steering Committee, International Conferences on Nonlinear Normal Modes and Localization, since 2012
- 3rd International Symposium "Rare Attractors and Rare Phenomena in Nonlinear Dynamics RA14, 2014, Riga - Jurmala, Latvia
- The 5th International conference on Nonlinear Normal Modes and Localization in Mechanics and Physics, Istanbul, Turkey, 2014
- The 6th International conference on Nonlinear Normal Modes and Localization in Mechanics and Physics, Liege, Belgium,, 2016
- EURODYN-10 conference, Rome, Italy, 2017

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- The 4th International Conference on vibro-impact-systems and systems with non-smooth interactions, Kassel, Germany, 30 July - 3 August 2018
- European Nonlinear Oscillations Conference Committee (ENOCC), 2021-2026
- **Organizer of CISM-AIMETA Advanced School on Exploiting the Use of Strong Nonlinearity in Dynamics and Acoustics, CC2206, 2022**

Organizer of Sessions and Symposia

- 7th EUROMECH Solid Mechanics Conference, Lisbon, September 7-11, 2009, Mini – Symposium: Nonlinear Localization and Targeted Energy Transfer in Dynamical Systems and Engineering
- 7th EUROMECH Nonlinear Dynamics Conference, Roma, Italy, 2011, Mini – Symposium 15: Energy transfer in Nonlinear Systems,
- International Conference on Vibration Problems, Section: Asymptotic Methods, Prague, 2011
- 8th EUROMECH Nonlinear Dynamics Conference, Vienna, Austria, 2014, Mini – Symposium: Energy transfer in Nonlinear Systems,
- Session organizer, ICTAM, Daegu, South Korea, 2024
- Session organizer, ICOVP, Lisbon, 2025

Teaching at special advanced professional courses by special invitations.

1. Advanced Nonlinear Strategies for Vibration Mitigation and System Identification, CISM, Udine, Italy, June 16-20, 2008
2. SICON – Marie Curie Training Course, event TC-4, Lyon, France, March 28-April 3, 2009
3. Modal Analysis of Nonlinear Mechanical Systems, CISM, Udine, Italy, June 25-29, 2012

Consultancy

1. Weizmann Institute of Science, Department of Chemical Physics.
2. Israel Institute of Metals, Technion.

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Reviewing for:

Physical Review Letters, Physical Review E, Applied Physics Letters, Nonlinear Dynamics, ASME Journal of Applied Mechanics, ASME Journal of Computational and Nonlinear Dynamics, Journal of Sound and Vibration, Journal of Vibration and Control, ASME Journal of Vibration and Acoustics, International Journal of Solids and Structures, Engineering Structures, International Journal of Impact Engineering, Meccanica, Europhysics Letters, International Journal of Non-Linear Mechanics, Journal of Physical Chemistry, Physica Status Solidi, Israel Journal of Chemistry, Entropy, Mathematical Reviews (AMS), Mechanic Research Communications, European Journal of Mechanics, Industrial & Engineering Chemistry Research, AIAA Journal , ASME and EUROMECH conferences, GIF, ISF